Sparse Innovation Regression with Compressed Sensing (SIR-CS)
Draft: Method and Synthetic Experiments

Abstract

We summarize a hybrid estimator that predicts a global component with machine learning and recovers a sparse innovation by compressed sensing under linear measurements. We then document a reproducible *Phase 0* synthetic baseline: ten independent seeds, measurement ratios up to $\rho = 0.6$, a refined ℓ_1 penalty grid for validation-based model selection, and standard diagnostics (curves versus ρ, gains over ml-only, parity plots, residual histograms, and example traces). A subsequent *Step 1* section compares FISTA versus SPGL1 on the same protocol. Real-data evaluation is outlined as future work.

1 Sparse Innovation Regression with Compressed Sensing (SIR-CS)

1.1 Problem setup

Let $u \in \mathbb{R}^p$ denote inputs and $y \in \mathbb{R}^N$ a high-dimensional target. At inference time we observe compressive measurements

$$b = My + \eta, \tag{1}$$

where $M \in \mathbb{R}^{m \times N}$ has $m \ll N$ and η models measurement noise.

We decompose the target into a learned background predictor and a residual:

$$y = f_\theta(u) + r, \tag{2}$$

where f_θ is a machine learning model and r is the modeling error. The structural assumption is that the innovation admits a sparse or compressible representation in a dictionary $\Psi \in \mathbb{R}^{N \times N}$:

$$r = \Psi\alpha^\star + \xi, \tag{3}$$

with $\alpha^\star$ sparse (or compressible) and ξ a small non-sparse mismatch term.

Substituting into the measurement model yields the reduced sensing model for the innovation coefficients:

$$z := b - Mf_\theta(u) = A\alpha^\star + \varepsilon, \qquad A := M\Psi, \qquad \varepsilon := M\xi + \eta. \tag{4}$$

Thus inference reduces to recovering $\alpha^\star$ from the linear measurements z rather than reconstructing all N coordinates of y directly.

1.2 Inference: recovery of the innovation

The basic estimator is the LASSO / BPDN problem

$$\hat{\alpha} \in \arg\min_{\alpha \in \mathbb{R}^N} \frac{1}{2}\|A\alpha - z\|_2^2 + \lambda\|\alpha\|_1, \tag{5}$$

followed by

$$\hat{y} = f_\theta(u) + \Psi\hat{\alpha}. \tag{6}$$

Structured variants. When prior support information, spatial regularity, or physics are available, we use

$$\hat{\alpha} \in \arg\min_{\alpha} \frac{1}{2}\|A\alpha - z\|_2^2 + \lambda_1\|W\alpha\|_1 + \lambda_2\,\Omega_{\text{str}}(\alpha) + \lambda_3\,\mathcal{P}(\Psi\alpha), \tag{7}$$

where W is a diagonal weight matrix (weighted ℓ_1), Ω_{str} encodes structured sparsity (e.g., group/tree penalties), and $\mathcal{P}$ encodes domain constraints (e.g., TV or physics-based penalties).

1.3 Optional learned support prior

Let g_ϕ predict support-related scores $\pi(u) \in [0,1]^N$. A simple weighting scheme sets, for thresholds $\tau \in (0,1)$ and $0 < \omega_{\text{low}} < 1$,

$$w_i = \begin{cases} \omega_{\text{low}}, & \pi_i(u) \geq \tau, \\ 1, & \text{otherwise}, \end{cases} \tag{8}$$

and solves weighted ℓ_1 with $\|W\alpha\|_1 = \sum_i w_i|\alpha_i|$.

1.4 Training objectives (two regimes)

Supervised training with sparse residual encouragement. When full y is available during training, one may train θ with

$$\mathcal{L}_{\text{train}} = \|f_\theta(u) - y\|_2^2 + \beta\|\Psi^\top(y - f_\theta(u))\|_1 + \gamma\,\Omega_{\text{str}}\Big(\Psi^\top(y - f_\theta(u))\Big), \quad (9)$$

when Ψ is orthonormal and analysis coefficients are appropriate; more generally, replace $\Psi^\top r$ by the analysis representation consistent with the chosen dictionary.

Task-aware training with simulated compression. Simulate $b = My + \eta$ during training and minimize end-to-end error after recovery,

$$\mathcal{L}_{\text{task}} = \|\hat{y}(u, b) - y\|_2^2, \quad (10)$$

using unrolled proximal methods or implicit differentiation if differentiability is required.

1.5 Informal stability statement (recovery stage only)

Proposition 1 (Informal)**.** *Assume $A = M\Psi$ satisfies a restricted isometry property of sufficient order for k-sparse vectors, and $\alpha^\star$ is k-compressible. Let $\hat{\alpha}$ be a BPDN/LASSO solution for noise level $\|\varepsilon\|_2$. Then typical stable recovery bounds imply*

$$\|\hat{\alpha} - \alpha^\star\|_2 \le C_1\|\varepsilon\|_2 + C_2\frac{\sigma_k(\alpha^\star)_1}{\sqrt{k}}, \quad (11)$$

and consequently

$$\|\hat{y} - y\|_2 \le \|\Psi\|_{2\to 2}\,\|\hat{\alpha} - \alpha^\star\|_2 + \|\xi\|_2. \quad (12)$$

1.6 Algorithm

1.7 Method variants (abbreviations)

V1 (SIR-CS): basic ℓ_1 on α. **V2 (WSIR-CS):** weighted ℓ_1 using W. **V3 (TSIR-CS):** temporal smoothing on α_t (e.g., $\|\alpha_t - \hat{\alpha}_{t-1}\|_2^2$). **V4 (Joint-SIR-CS):** shared α_c and channel-specific α_j across multiple outputs/sensors.

Algorithm 1 SIR-CS (inference)

Require: Measurements b, operator M, model f_θ, dictionary Ψ, weights W (optional), penalties $\Omega_{\text{str}}, \mathcal{P}$ (optional)

$\hat{y}_{\text{bg}} \leftarrow f_\theta(u)$
$z \leftarrow b - M\hat{y}_{\text{bg}}$
if support prior available **then**
  $\pi \leftarrow g_\phi(u)$; build $W(\pi)$
end if
$\hat{\alpha} \leftarrow \arg\min_\alpha \frac{1}{2}\|M\Psi\alpha - z\|_2^2 + \lambda_1\|W\alpha\|_1 + \lambda_2\Omega_{\text{str}}(\alpha) + \lambda_3\mathcal{P}(\Psi\alpha)$
$\hat{y} \leftarrow \hat{y}_{\text{bg}} + \Psi\hat{\alpha}$

Ensure: $\hat{y}$

2 Experiments: synthetic validation (Phase 0 baseline)

This section reports a *Phase 0* controlled synthetic study: a frozen protocol aligned with the project's ablation roadmap. The goal is to verify that the SIR-CS pipeline behaves consistently with Section 1 in a favorable regime (learnable global component, sparse innovation in the recovery basis, Gaussian compressive sensing). Uncertainty is quantified across *independent* random seeds (not pseudo-replications over test samples). A complementary evaluation on *real data* remains future work (Section 2.8).

2.1 Protocol

Phase 0 reference configuration. We fix the data-generating process and training protocol, then aggregate statistics across ten seeds $\{7, 13, 23, 29, 31, 37, 41, 43, 47, 53\}$. Measurement ratios are $\rho \in \{0.20, 0.30, 0.40, 0.50, 0.60\}$ with $N = 128$ and $m = \lfloor \rho N \rfloor$. All other hyperparameters match the implementation defaults for this profile: $p = 12$, $k = 6$, $\Psi = I$, `residual_mode`=`support_from_u`, $Gaussian$M, `measurement_noi`

Data generation. Inputs $u \in \mathbb{R}^p$ are drawn i.i.d. standard Gaussian. Targets satisfy $y = y_{\text{bg}} + \Psi\alpha^\star + \xi_{\text{out}}$ where y_{bg} is a smooth nonlinear random-feature map of u, $\alpha^\star$ is k-sparse in canonical coordinates, $\Psi = I$, and ξ_{out} is small output noise.

Train and test splits. We use $n_{\text{train}} = 1200$, $n_{\text{val}} = 300$, and $n_{\text{test}} = 300$ samples per seed.

Measurement model. Observations are $b = My + \eta$ with additive Gaussian measurement noise at fixed standard deviation, shared consistently between validation-based λ selection and test evaluation in the code.

Methods. **ml_only:** multi-output MLP predicting y from u without using M at inference. **hybrid:** SIR-CS with FISTA/LASSO on the innovation after subtracting $Mf_\theta(u)$ from b. **weighted_hybrid:** same as hybrid with diagonal weights in the ℓ_1 term derived from an auxiliary predictor of coefficient magnitudes. **cs_only:** sparse reconstruction of y in Ψ from b without the learned background (misspecified reference).

Hyperparameters and model selection. The ℓ_1 penalty is selected on the validation set over the grid $\{10^{-4}, 3\cdot10^{-4}, 10^{-3}, 3\cdot10^{-3}, 10^{-2}, 3\cdot10^{-2}\}$. Lipschitz constants for FISTA are precomputed once per (M, Ψ). Reported numbers aggregate *first* over test samples within each (seed, ρ), then *across seeds* as mean $\pm$ standard deviation.

2.2 Main quantitative results

Table 1 summarizes test RMSE and MAE. The ml-only baseline is *flat* in ρ because its prediction does not use the realized compressive measurements at test time.

Reading Table 1. `hybrid` achieves the lowest RMSE and MAE at every ρ. The margin over ml-only grows monotonically with ρ in this table: RMSE drops from 0.369 to 0.245 at $\rho = 0.60$ (absolute reduction 0.124 relative to ml-only at the same ρ). `cs_only` remains far worse because it omits the global component $f_\theta(u)$. `weighted_hybrid` improves consistently over ml-only but remains above `hybrid`; the RMSE gap `weighted_hybrid`−`hybrid` widens slightly as ρ increases, which motivates dedicated prior diagnostics (support correlation, oracle weights) rather than treating the weighted variant as the primary estimator in this regime.

2.3 Error versus measurement ratio

Figure 1 shows RMSE, MAE, and relative ℓ_2 error versus ρ. Error bars show the standard deviation across seeds.

Table 1: Test RMSE and MAE (mean ± std across ten seeds) versus $\rho = m/N$. Best RMSE in each row in bold (excluding `cs_only`).

ρ	`ml_only`	`hybrid`	`weighted_hybrid`	`cs_only`
RMSE				
0.20	0.369 ± 0.011	$\mathbf{0.336} \pm 0.011$	0.340 ± 0.011	0.625 ± 0.015
0.30	0.369 ± 0.011	$\mathbf{0.319} \pm 0.011$	0.325 ± 0.011	0.593 ± 0.016
0.40	0.369 ± 0.011	$\mathbf{0.296} \pm 0.011$	0.305 ± 0.011	0.553 ± 0.015
0.50	0.369 ± 0.011	$\mathbf{0.272} \pm 0.011$	0.283 ± 0.010	0.508 ± 0.013
0.60	0.369 ± 0.011	$\mathbf{0.245} \pm 0.010$	0.258 ± 0.010	0.455 ± 0.015
MAE				
0.20	0.283 ± 0.011	$\mathbf{0.262} \pm 0.010$	0.266 ± 0.010	0.500 ± 0.012
0.30	0.283 ± 0.011	$\mathbf{0.251} \pm 0.010$	0.256 ± 0.010	0.475 ± 0.013
0.40	0.283 ± 0.011	$\mathbf{0.234} \pm 0.009$	0.241 ± 0.009	0.443 ± 0.012
0.50	0.283 ± 0.011	$\mathbf{0.215} \pm 0.009$	0.225 ± 0.009	0.407 ± 0.011
0.60	0.283 ± 0.011	$\mathbf{0.194} \pm 0.008$	0.205 ± 0.008	0.364 ± 0.012

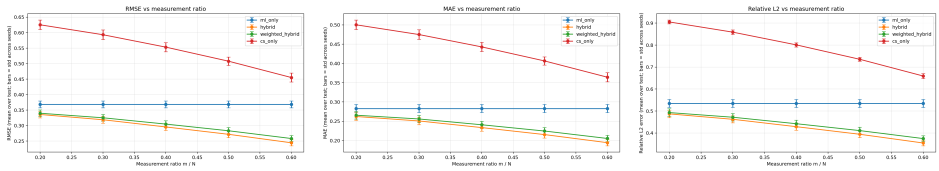

Figure 1: Test error versus $\rho = m/N$ for four methods (Phase 0). Bars: std across ten seeds. ml-only is flat in ρ by construction.

Discussion. The monotone improvement of `hybrid` with ρ matches the inverse-problem interpretation in Section 1: more linear measurements improve conditioning of the effective operator $M\Psi$ at fixed noise. The flat ml-only curve is expected: the baseline ignores M at prediction time. Relative ℓ_2 tracks RMSE/MAE: `cs_only` saturates at high error, while hybrid methods follow the same ordering as in Table 1.

2.4 Structured recovery metrics and grouped RMSE

Figure 2 (left) reports support F1 for methods that yield sparse coefficients comparable to ground truth. `ml_only` is omitted. Figure 2 (right) shows grouped RMSE by ρ.

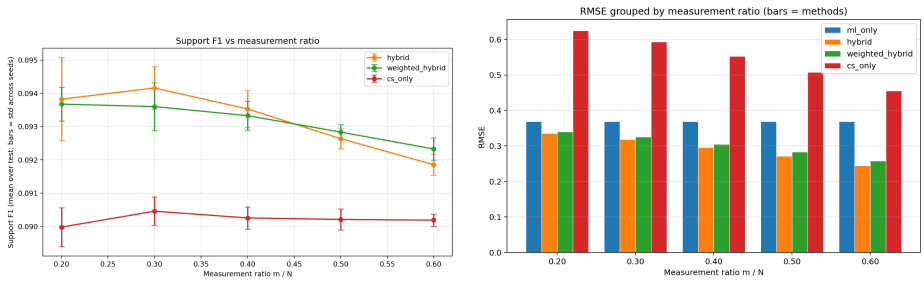

Figure 2: Left: support F1 (Phase 0). Right: grouped RMSE by ρ.

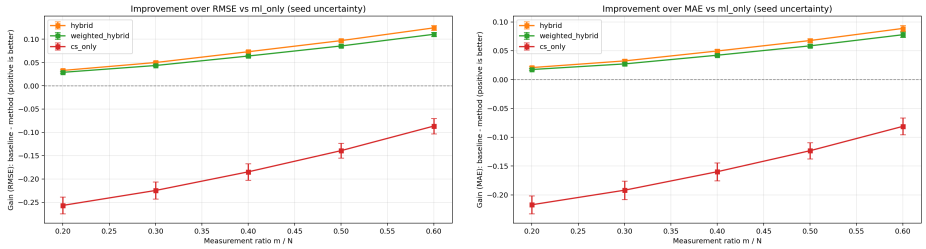

Figure 3: Gain over ml-only (positive is better). `hybrid` and `weighted_hybrid` are positive at all ρ; `cs_only` is negative throughout.

Discussion. Support F1 stays in a narrow band for all sparse methods, with hybrid methods slightly above `cs_only`; this metric should be read jointly with reconstruction error, because high F1 alone does not rescue `cs_only` under misspecification. The bar panel mirrors Table 1: decreasing hybrid error with ρ, flat ml-only, and elevated `cs_only`.

2.5 Gains over the ml-only baseline

Figure 3 shows per-seed gains $\mathrm{RMSE}_{\mathrm{ml}} - \mathrm{RMSE}_{\mathrm{method}}$ (and MAE analog), aggregated across seeds. Positive values mean improvement over ml-only.

Discussion. `hybrid` and `weighted_hybrid` yield uniformly positive gains in this Phase 0 window, with larger gains at higher ρ. `hybrid` dominates `weighted_hybrid` in both panels, consistent with Table 1. `cs_only` remains strongly negative, quantifying the mismatch when the dense background is ignored.

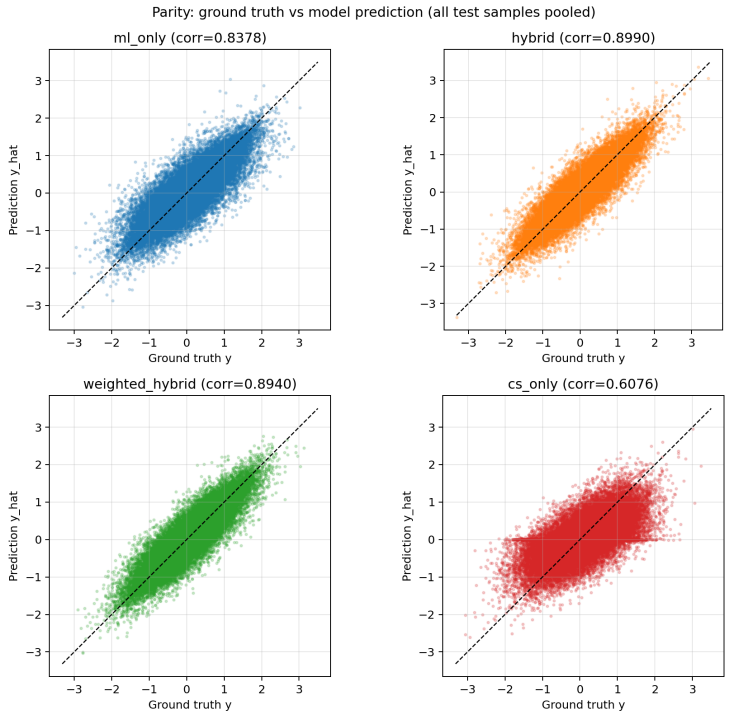

Figure 4: Parity plots (Phase 0): ground truth versus prediction. The identity line is dashed. `hybrid` aligns most tightly with identity; `cs_only` shows a horizontal band near $\hat{y} \approx 0$ for a range of true values.

2.6 Parity plots, residuals, and example traces

Figure 4 pools test coordinates across seeds and ρ. Figure 5 shows histograms of $\hat{y} - y$. Figure 6 shows example traces along the output index.

Discussion. Parity plots provide a global calibration check: `hybrid` tracks the identity line most closely, while `cs_only` exhibits the classic artifact of a misspecified sparse model under dense targets. Residual histograms and traces reinforce the quantitative ordering in Table 1.

2.7 Limitations

These results establish internal consistency and a favorable synthetic regime. They do *not* guarantee performance under arbitrary real-world conditions (non-Gaussian M, basis mismatch, ill-conditioned subsampling operators,

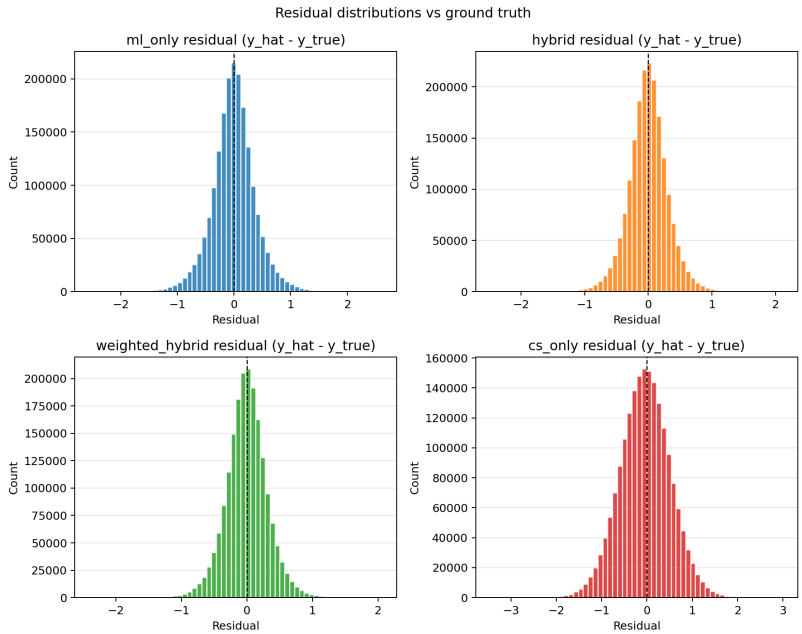

Figure 5: Residual histograms $\hat{y} - y$ (Phase 0). Hybrid methods concentrate more mass near zero than ml-only; `cs_only` is heavy-tailed.

heavy-tailed noise, or poorly calibrated λ grids). In validation logs for Phase 0, the selected λ often lies at the *smallest* grid value (10^{-4}), suggesting that future sweeps could extend the grid toward smaller penalties if computational budget allows.

2.8 Outlook: evaluation on real data

The next experimental milestone is to instantiate the same comparison template on a real dataset with physically motivated M and Ψ (for example, a field subsampled at sensor locations). The Phase 0 section then serves as a reproducible sanity check and ablation reference, while the real-data study targets external validity and domain baselines.

3 Experiments: classical sparse solvers (Step 1)

This section documents a *Step 1* comparison aligned with the project roadmap: the sparse recovery block is run with two classical solvers, FISTA (penalized

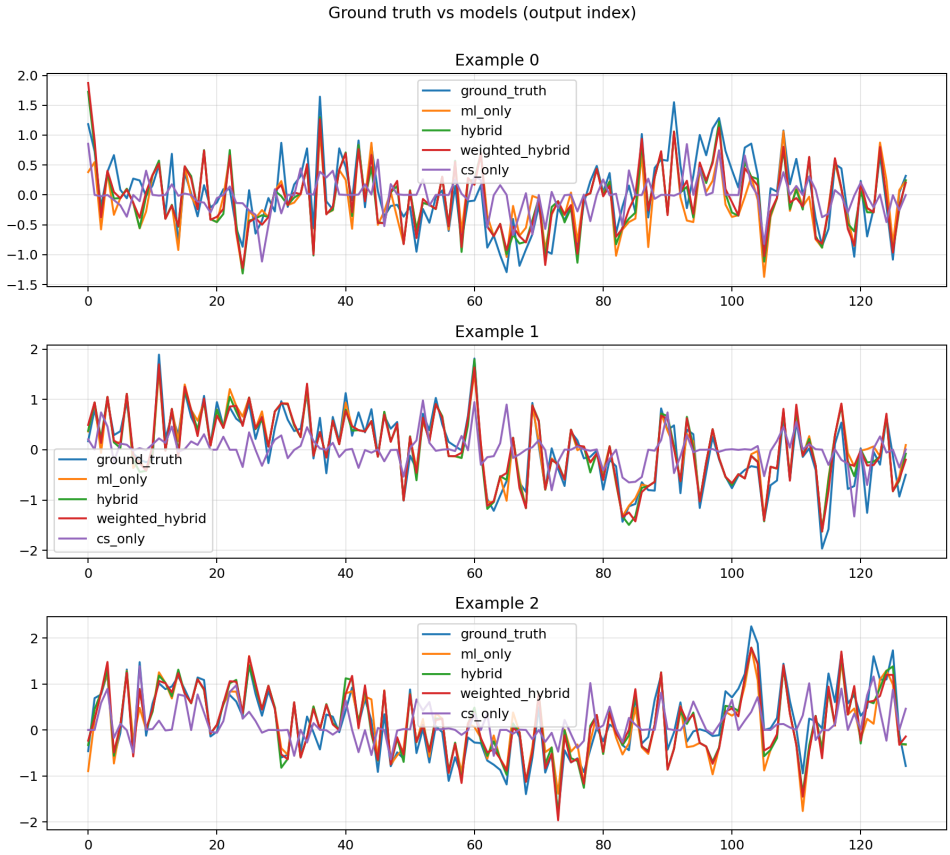

Figure 6: Example reconstructions along output index (Phase 0). ML-based methods track the ground truth; `cs_only` deviates toward an overly sparse profile.

ℓ_1) and SPGL1 (LASSO mode via PyLops), while holding the Phase 0 data protocol fixed. The goal is to separate formulation quality from optimizer choice before exploring trainable solvers or external baselines.

3.1 Protocol

We reuse the Phase 0 configuration in Section 2.1: ten seeds $\{7, 13, 23, 29, 31, 37, 41, 43, 47, 53\}$, $\rho \in \{0.20, 0.30, 0.40, 0.50, 0.60\}$, $N = 128$, $m = \lfloor \rho N \rfloor$, Gaussian M, identity Ψ, and the same noise levels. The implementation profile is `solver_comparison` with `dual_cs_solver=True`, which evaluates `hybrid_fista`, `hybrid_spgl1`, `cs_only_fista`, `cs_only_spgl1`, and `ml_only` (weighted hybrid is disabled

Table 2: Test RMSE (mean ± std across ten seeds) versus $\rho = m/N$ for Step 1. Frozen run `20260417_114526`. Best hybrid/cs row in each block: bold among `hybrid_*` and `cs_only_*` (lowest RMSE).

ρ	`ml_only`	`hybrid_fista`	`hybrid_spgl1`	`cs_only_fista`	`cs_only_spgl1`
0.20	0.369 ± 0.011	$\mathbf{0.336 \pm 0.011}$	0.368 ± 0.011	0.625 ± 0.015	0.690 ± 0.013
0.30	0.369 ± 0.011	$\mathbf{0.319 \pm 0.011}$	0.367 ± 0.011	0.593 ± 0.016	0.690 ± 0.013
0.40	0.369 ± 0.011	$\mathbf{0.296 \pm 0.011}$	0.367 ± 0.011	0.553 ± 0.015	0.689 ± 0.013
0.50	0.369 ± 0.011	$\mathbf{0.272 \pm 0.011}$	0.367 ± 0.011	0.508 ± 0.013	0.689 ± 0.013
0.60	0.369 ± 0.011	$\mathbf{0.245 \pm 0.010}$	0.367 ± 0.011	0.455 ± 0.015	0.689 ± 0.013

in this profile).

Sparse optimization. FISTA selects λ on the validation grid $\{10^{-4}, 3 \cdot 10^{-4}, 10^{-3}, 3 \cdot 10^{-3}, 10^{-2}, 3 \cdot 10^{-2}\}$. SPGL1 runs in LASSO mode ($\tau > 0$, $\sigma = 0$) with $\tau \in \{10^{-4}, 3 \cdot 10^{-4}, 10^{-3}, 3 \cdot 10^{-3}, 10^{-2}, 3 \cdot 10^{-2}, 10^{-1}\}$. Software: `pylops` matrix operator with dense $A = M\Psi$ and the `spgl1` Python package required by the PyLops wrapper.

Artifacts. The frozen run used for tables and figures in this draft is `outputs/solver_comparison/runs/20260417_114526/` (CSVs, `PROTOCOL.txt`, logs). Figures are copied under `paper/figures/solver_comparison/runs/20260417_114526/` with filenames `01_...10_` matching the Phase 0 dashboard.

3.2 Main quantitative comparison

Table 2 reports test RMSE (mean ± std across seeds). The best hybrid row at each ρ is in bold; `cs_only_*` rows are excluded from bolding.

Reading Table 2. `hybrid_fista` matches the Phase 0 `hybrid` behavior: it improves monotonically with ρ and beats `ml_only` at every ratio. `hybrid_spgl1` is essentially *flat* near the ml-only RMSE (≈ 0.367) across ρ, so the SPGL1 LASSO instantiation tested here does not realize the hybrid gain seen under FISTA. `cs_only_fista` improves with ρ but remains far above hybrid methods; `cs_only_spgl1` stays near 0.69 with almost no ρ sensitivity, consistent with failed reconstruction (relative ℓ_2 near one in the logs).

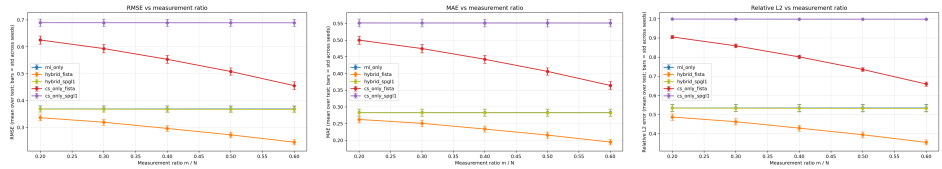

Figure 7: Step 1: RMSE, MAE, and relative ℓ_2 versus ρ for FISTA/SPGL1 variants. Error bars: std across seeds. `ml_only` is flat in ρ.

3.3 Figures

Figure 7 mirrors the Phase 0 dashboard for Step 1 methods (curves versus ρ, gains, parity, residuals, traces). PNG assets for this run live in the repository tree `paper/figures/solver_comparison/runs/` with the same `run_id` as in Table 2.

3.4 Takeaway

Under this frozen synthetic protocol, **FISTA remains the preferred classical sparse solver** for the hybrid and `cs_only` baselines. The tested SPGL1 configuration does not improve hybrid error relative to FISTA; the hybrid SPGL1 curve is numerically close to ml-only, while `cs_only_spgl1` is strongly degraded. This supports keeping FISTA as the default classical block in subsequent roadmap steps (trainable solvers, robustness sweeps) unless a revised SPGL1 parameterization or scaling study changes the conclusion.

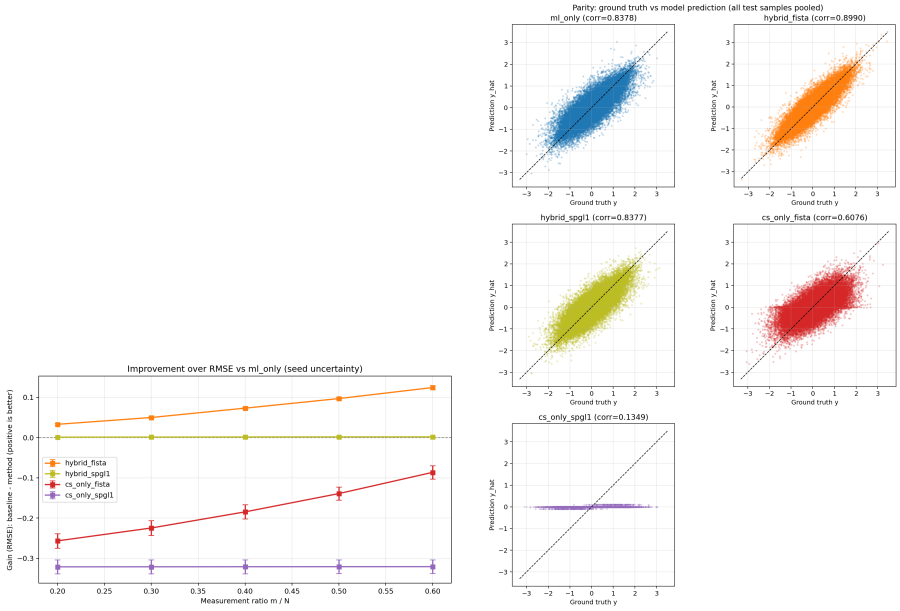

Figure 8: Step 1: RMSE gain over ml-only (left) and parity plot (right).